



Society of Petroleum Engineers

SPE-226905-MS

The Use of Projection-Based Model Order Reduction for Efficient Simulation of Subsurface Carbon Storage Via Adaptive and Non-Adaptive Sampling Techniques

A. H. Alali and M. M. Alsinan, Energy Sciences and Engineering, Stanford University, Stanford, California, USA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226905-MS](https://doi.org/10.2118/226905-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Conducting a comprehensive risk and uncertainty analysis for a Geologic CO₂ Storage (GCS) project typically requires simulating numerous realizations to ensure thorough coverage of expected design parameter ranges. Due to the complexity of these models and the need for long simulation periods to reach steady-state conditions and verify CO₂ containment, high-performance computing (HPC) resources are commonly employed. Given the computational cost and demands, alternative innovative techniques are essential to streamline this process.

In this study, we employ Projection-Based Model Order Reduction (PMOR) to evaluate the accuracy of three sampling techniques including Random, Latin Hypercube, and Greedy sampling, as compared to a High-Fidelity Model (HFM). The objective is to minimize the number of simulation runs while maintaining accuracy. The case study considers multiple simulation scenarios involving CO₂ injection at varying rates and cumulative times within an aquifer, of a field characterized by two key uncertain attributes: rock permeability and rock porosity, which govern the reservoir flow and storage capacity. The HFM used is based on multiple 2D sections with 60 × 220 cell blocks from the 10th SPE Comparative Solution Project.

Each cell in the model has two degrees of freedom, pressure and saturation, totaling 26,400 degrees of freedom. The HFM simulations were conducted using the open-source simulator GEOS, leveraging its subsurface flow simulation capabilities while excluding flow-poromechanics and thermal coupling. All simulations were executed on Stanford's HPC cluster. The study demonstrates that applying a linear Petrov-Galerkin method in the upper heterogeneous and isotropic layers of the model can reduce the number of degrees of freedom significantly, achieving less than 5% error with just 11 non-adaptive samples, and further reducing the required samples to five with adaptive sampling. However, the lower anisotropic layers exhibit greater geological complexity, necessitating more advanced nonlinear model order reduction methods to accurately capture their impact on reservoir characteristics.

Over the years, subsurface simulation research has primarily focused on oil and gas applications, with many of these methods subsequently adapted for GCS modeling. However, the computational cost of GCS simulations has led to the adoption of probabilistic models as an alternative. However, due to the high uncertainty, large number of simulations and realizations are required for this approach. More

recently, machine learning and deep learning have been explored as data-driven approaches, both physics-agnostic and physics-informed, to approximate solutions without solving the full governing equations. In contrast, this study revisits solving the governing physics, leveraging surrogate models projected onto lower-dimensional spaces to retain accuracy while achieving exceptional computational efficiency. Our findings pave the way for extending these techniques to more complex, fully coupled simulations that integrate all relevant physics governing CO₂ storage.

Introduction

Carbon capture and geological sequestration, which is the storage of CO₂ in underground formations, has become increasingly recognized as a vital strategy in the global effort to address climate change and reach net-zero emissions. Carbon dioxide, a major greenhouse gas, plays a central role in climate change. Extensive research has shown that rising atmospheric CO₂ levels are directly linked to increasing global temperatures, a trend expected to continue throughout this century without significant intervention [Houghton, 1996; Wigley et al., 1996; Hoffert et al., 1998; Oreskes, 2004; Farnsworth and Lichter, 2012; Cook et al., 2013; Zhai et al., 2018; Myers et al., 2021]. Given that over 80% of the world's primary energy still comes from fossil fuels [BP, 2021], which is a major source of CO₂ emissions, the current shift toward renewable energy alone is not enough to bring about the necessary emissions reductions. Therefore, additional fast-acting and effective solutions are essential.

Geological carbon storage is emerging as a promising complementary approach to help mitigate climate change. This technique involves injecting supercritical CO₂ deep underground, where it can remain permanently trapped, preventing its release into the atmosphere [Hitchon et al., 1999; Orr Jr, 2004; Metz et al., 2005; Townsend et al., 2020]. Alongside other environmental initiatives, this technology supports the broader goal of developing a sustainable future within the coming decades. Among the various geological storage options, saline aquifers are the most widely used [Bruant et al., 2002; Pruess et al., 2003].

To ensure the effectiveness and safety of CO₂ storage, simulation studies play a critical role in predicting how CO₂ behaves after injection. These models inform project design and economic assessments by estimating how much CO₂ can be securely stored without risk of leakage. A key goal of such studies is to quantify the volume of CO₂ that becomes permanently trapped, maximizing storage efficiency while maintaining containment within the designated geological formations. Achieving this is crucial, as it determines the potential contribution of each storage project toward global carbon reduction efforts and meeting the net-zero targets established under the 2015 Paris Agreement.

Objective and Brief Description

The main objective of this study is to propose a workflow for conducting a comprehensive risk and uncertainty analysis for a CO₂ storage project through subsurface modeling by the use of model reduction. This workflow relies on applying Projection-Based Model Order Reduction (PMOR) and conclude by comparing the accuracy of results across three sampling techniques: Random, Latin Hypercube, and Greedy sampling. The case study focuses on a project that involves CO₂ injection into the aquifer of a field characterized by key uncertain attributes (among other attributes) which are: rock permeability, and rock porosity. The quantity of interest (QoI) assessed is storage capacity or total injected volume at steady-state conditions (which sometimes can be on the scale of 100's of years), a key target for each of these projects. This QoI can be assessed by two subsurface dynamic properties which are: pore pressure, and CO₂ content (or saturation). The pore pressure assessment ensures the injection of CO₂ will not exceed certain limit preventing any leakage into unwanted territories due to rock failure from the overburden or under-burden. On the other hand, the saturation assessment ensures the containment of the CO₂ plume after injection within the field. These two parameters are typical outputs in (HDM) predictions. These HDM simulations are

performed using the open-source simulator (GEOS) [Settgast et al., 2024] here by relying on its subsurface flow simulation capabilities while neglecting the flow-poromechanics and thermal coupling capabilities for the purposes of this study. Also, the high-fidelity simulations were carried out in Stanford Sherlock HPC cluster by making use of the MPI and GPU capability of the simulator to reduce the computational time. However, the goal is to propose a workflow that rely on PMOR that will allow greater reductions in computational times for such problems.

Geological Model

We are utilizing the 10th SPE comparative project model, which is commonly used as a benchmark for upscaling methods. It is a 3-D geological model that consists of $60 \times 220 \times 85$ grid cells, each of size $20\text{ft} \times 10\text{ft} \times 2\text{ft}$. The geological model represents the Jurassic Upper Brent formations that is common in many North Sea fields. Also, in this model, the top 70 ft (35 layers) represent the shallow-marine Tarbert formation which is characterized by heterogeneous but isotropic porosity and permeability fields. On the other hand, the lower 100 ft (50 layers) represent the fluvial Ness formation, which is characterized by high permeability channels that results in heterogeneous and anisotropic permeability and porosity fields. Figures 1 and 2 show distribution of porosity and horizontal permeability in the Tarbert and Ness layers, which demonstrate a very rich parameter space. Also, we expect that differences between the parameters of Tarbert and Ness formations would result in different flow patterns. Specifically, high-permeability channels in Ness formation would result in a more pronounced CO_2 viscous-dominated flow compared to Tarbert, where the flow is more gravity- and/or capillary-dominated.

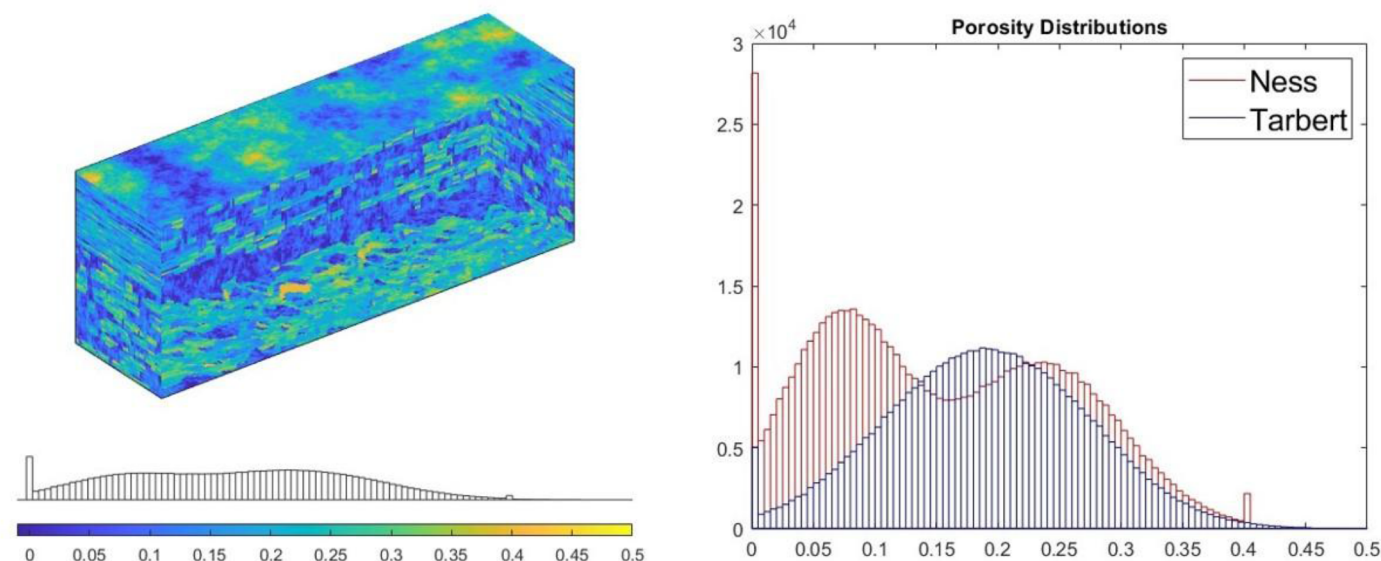


Figure 1—Geological description: Porosity distribution.

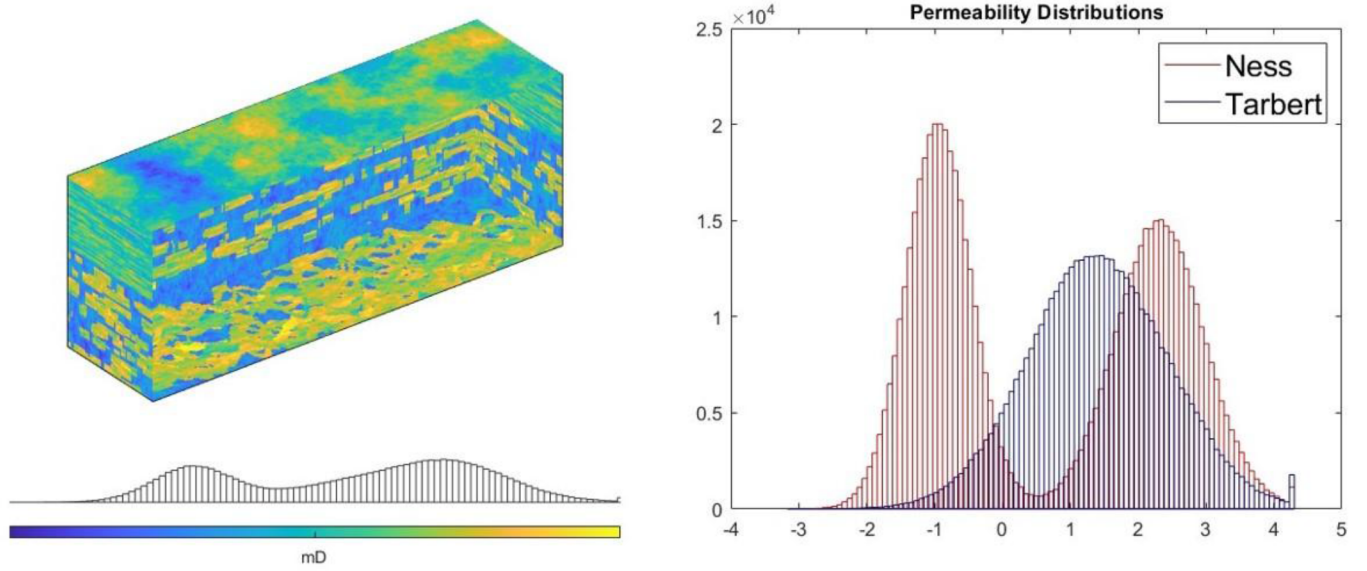


Figure 2—Geological description: Permeability distribution.

Mathematical Model

The problem of interest is related to multiphase flow in porous media under the influence of viscous, gravity, and capillary forces. This can be expressed as a flow and transport problem as explained in the next sections.

Governing Equations

We consider a two-component system of gas and water flow in a slightly compressible porous medium, in which both components can exist only in their corresponding phases of vapor and liquid brine. The gas and water components are denoted by the subscripts g and w , respectively, and the liquid, which is the wetting phase, and the vapor, the non-wetting phase, are denoted by the subscripts ℓ and v , respectively. The mass conservation laws are

$$\frac{\partial}{\partial t}(\phi \rho_v S_v) + \nabla \cdot (\rho_v \mathbf{u}_v) = \rho_v q_v, \quad (1)$$

and

$$\frac{\partial}{\partial t}(\phi \rho_\ell S_\ell) + \nabla \cdot (\rho_\ell \mathbf{u}_\ell) = \rho_\ell q_\ell \quad (2)$$

where $\phi(\mathbf{x}, p)$ is the porosity of the medium which is a function of pressure, $S_{\ell(\mathbf{x}, t)}$ is the saturation of the phase ℓ and similarly for the phase v , and t is the time. The source/sink terms q_ℓ and q_v are positive for injection and negative for production. The phase velocity, $\mathbf{u}_\ell$ and $\mathbf{u}_v$, are defined using the multiphase extension of Darcy's law as

$$\mathbf{u}_\ell := -k\lambda_\ell(\nabla p_\ell - \rho_\ell g \nabla z), \quad (3)$$

and

$$\mathbf{u}_v := -k\lambda_v(\nabla p_v - \rho_v g \nabla z), \quad (4)$$

Here, $k(x)$ is the scalar absolute permeability of the medium, λ_ℓ is the phase mobility of the liquid phase defined as $k_{r\ell}/\mu_\ell$, where $k_{r\ell}(x, S_\ell)$ is the phase relative permeability, μ_ℓ is the phase viscosity, and ρ_ℓ is the phase density. These are also defined similarly for the vapor phase. The gravitational acceleration is denoted by g , and the depth by z (positive going downward). The conservation of mass equations are constrained by the following volume constraint:

$$S_\ell + S_v = 1. \quad (5)$$

Moreover, the capillary pressure constraint relates both phase pressures with

$$P_c(S_\ell) = p_v - p_\ell \quad (6)$$

which closes the system for the problem of interest.

Flow and Transport Equation

In this section we show how the problem of interest can be re-written to be divided into a flow and a transport equation, and with this we will be able to show the coupling between the different physical processes. More specifically, this formulation is obtained by decomposing the governing equations into a flow problem for both phases and a transport problem for one of the two phases. To obtain this decomposition, we first sum the phase velocities to define a total velocity, and since the flow is compressible with variable density, we rely on a total mass flux as:

$$U_T := \rho_\ell \mathbf{u}_\ell + \rho_v \mathbf{u}_v = -k(\rho_\ell \lambda_\ell + \rho_v \lambda_v) \nabla p + k(\lambda_\ell \rho_\ell^2 + \lambda_v \rho_v^2) g \nabla z + k \rho_\ell \lambda_\ell \nabla P_c \quad (7)$$

This total mass flux is a function of space as well as of the reference pressure chosen to be $p = p_v$ and the reference saturation denoted by $S = S_\ell$.

Now by summing (1) and (2) then using the mass constraint and the definition of the total mass flux, this results in the following elliptic PDE governing the temporal evolution of the pressure field which is known as the flow equation:

$$\frac{\partial}{\partial t}(\phi \rho_t) + \nabla \cdot (\rho_\ell \mathbf{u}_\ell + \rho_v \mathbf{u}_v) = \rho_\ell q_\ell + \rho_v q_v \quad (8)$$

where

$$\rho_t = \rho_\ell S_\ell + \rho_v S_v \quad (9)$$

Next, the highly nonlinear parabolic transport equation is obtained by using (7) to formally eliminate the pressure variable from (1) and (2), yielding

$$\frac{\partial}{\partial t}(\phi \rho_v S_v) + \nabla \cdot F_v = \rho_v q_v \quad (10)$$

and

$$\frac{\partial}{\partial t}(\phi \rho_\ell S_\ell) + \nabla \cdot F_\ell = \rho_\ell q_\ell \quad (11)$$

where the flow flux for each phase is defined as

$$F_\ell := \frac{\rho_\ell \lambda_\ell}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} U_T + k \frac{\rho_\ell \lambda_\ell \rho_v \lambda_v}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} (\rho_\ell - \rho_v) g \nabla z + k \frac{\rho_\ell \lambda_\ell \rho_v \lambda_v}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} (\nabla P_c) \quad (12)$$

and

$$F_v := \frac{\rho_v \lambda_v}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} U_T + k \frac{\rho_\ell \lambda_\ell \rho_v \lambda_v}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} (\rho_v - \rho_\ell) g \nabla z - k \frac{\rho_\ell \lambda_\ell \rho_v \lambda_v}{\rho_\ell \lambda_\ell + \rho_v \lambda_v} (\nabla P_c), \quad (13)$$

where both transport equations are comprised of three terms that correspond to the viscous, gravity, and capillary forces as stated earlier.

With this, the PDE's can be discretized using a proper numerical method prior to solving them.

Numerical Model

To solve the highly nonlinear system of PDE's, we resort to numerical methods to simulate the flow and transport in porous media of subsurface formations by discretizing them in space and time using appropriate techniques as explained next.

Discretization of the Governing Equations

Here we choose to use a backward Euler in time and finite-volume discretization in space of the two components' mass conservation equations (1) and (2) for a grid of arbitrary structure as follows:

$$V_i \frac{(\phi \rho_\ell S_\ell)_i^{n+1} - (\phi \rho_\ell S_\ell)_i^n}{\Delta t} + \sum_{j \in \text{ad } j(i)} (\rho_\ell f_{\ell ij})^{n+1} = V_i (\rho_\ell \tilde{q}_\ell)_i^{n+1}, \quad \forall i \in \{1, \dots, N\} \quad (14)$$

$$V_i \frac{(\phi \rho_v S_v)_i^{n+1} - (\phi \rho_v S_v)_i^n}{\Delta t} + \sum_{j \in \text{ad } j(i)} (\rho_v f_{vij})^{n+1} = V_i (\rho_v \tilde{q}_v)_i^{n+1}, \quad \forall i \in \{1, \dots, N\} \quad (15)$$

where the densities in the above equation are obtained via upstream weighting, as will be described subsequently, while the volumetric flow between cells are defined as

$$f_{vij} = T_{ij} \lambda_{vij} \Delta \Phi_{vij} \quad (16)$$

and

$$f_{\ell ij} = T_{ij} \lambda_{\ell ij} \Delta \Phi_{\ell ij} \quad (17)$$

where T_{ij} is the geometric transmissibility between cells, determined from the geometric average of the one-sided transmissibilities, denoted by as $\hat{T}_{ij}$

$$T_{ij} = T_{ji} = \left(\widehat{T_{ij}^{-1}} + \widehat{T_{ji}^{-1}} \right)^{-1} = \left(\frac{d_{i(ij)}}{k_i A_{ij}} + \frac{d_{j(ji)}}{k_j A_{ji}} \right)^{-1} \quad (18)$$

in which k_i (respectively, k_j) is the rock absolute permeability in cell i (respectively, j) in the direction of flow, $d_{i(ij)}$ (respectively, $d_{j(ji)}$) is the distance from the center of cell i (respectively, j) to the edge of the cell, and $A_{ij} = A_{ji}$ is the area of the edge of the cell. In addition, the difference in phase potentials between neighboring cells are defined as

$$\Delta \Phi_{vij} = \Phi_{vi} - \Phi_{vj} = \Delta p_{ij} - \bar{\rho}_v g \Delta z_{ij}, \quad (19)$$

and

$$\Delta \Phi_{\ell ij} = \Phi_{\ell i} - \Phi_{\ell j} = \Delta p_{ij} - \bar{\rho}_\ell g \Delta z_{ij} - (\Delta P_c)_{ij} \quad (20)$$

Where $\bar{\rho}_v$ and $\bar{\rho}_\ell$ represent the arithmetic average of the densities of cells i and j . For the standard Phase-Potential Upwinding (PPU) scheme, [Aziz and Settari, 1979], the phase mobilities at an interface (ij) between the two cells are approximated using the upstream values of our primary unknowns, as such for the vapor phase here

$$\lambda_{vij} = \begin{cases} \lambda_v(p_{vi}, S_i) & \text{if } \Delta \Phi_{vij} \geq 0 \\ \lambda_v(p_{vj}, S_j) & \text{otherwise} \end{cases} \quad (21)$$

Numerical Solution using Fully Implicit Scheme

For the backward Euler fully-implicit solution procedure of the discretized equations we first cast both conservation equations in residual form and date every unknown and parameter at the latest time-step

$$R_\ell = V_i \frac{(\phi \rho_\ell S_\ell)_i^{n+1} - (\phi \rho_\ell S_\ell)_i^n}{\Delta t} + \sum_{j \in \text{ad } j(i)} (\rho_\ell f_\ell)_{ij}^{n+1} - V_i (\rho_\ell \tilde{q}_\ell)_i^{n+1} = 0, \quad \forall i \in \{1, \dots, N\} \quad (22)$$

$$R_v = V_i \frac{(\phi \rho_v S_v)_i^{n+1} - (\phi \rho_v S_v)_i^n}{\Delta t} + \sum_{j \in \text{ad } j(i)} (\rho_v f_v)_{ij}^{n+1} - V_i (\rho_v \tilde{q}_v)_i^{n+1} = 0, \quad \forall i \in \{1, \dots, N\} \quad (23)$$

Then, we linearize them and solve for the pressure and saturation of each cell using Newton's method

$$J^k \delta X = -R^k \quad (24)$$

So that

$$X^{k+1} = \delta X + X^k \quad (25)$$

Generating HDM Solutions

To streamline running 765 simulations of the high-dimensional model solutions, we relied on Stanford Sherlock high-performance computing cluster by making advantage of the GPU and parallel computing features. The main outcome of these simulation runs is to store solutions, associated Jacobians and right-hand side residual vectors. Each simulation case requires inputs on rock, fluid and well properties. The rock properties were obtained from the static model whereas the fluid properties were obtained from NIST database, which is summarized in the table below. For the boundary conditions, all cases include a CO₂ injector in the middle of the 2-D section (which entails a source flux term) and four Dirichlet boundary conditions (i.e. sink terms with fixed pressure) on the corners of the domain.

Table 1—Summary of initial Conditions

Water Saturation (%)	100.0
Pressure (Pa)	4.134×10^7

Table 2—Summary of input fluid properties

Component	Density (kg/m ³)	Viscosity (Pa.s)
CO ₂	368.13	3.80×10^{-5}
Water	764.36	9.60×10^{-4}

Parameter Selection

Since the objective of developing the reduced-order model (ROM) is to capture the injected CO₂ plume at different times, we selected four parameters that have known significant impact on the plume migration, which are porosity, horizontal permeability, injection rate, and total injection time. We neglected vertical permeability since we are interested in predicting flow in 2-D sections of the original model, in which case only horizontal permeability is relevant. Figure 3 shows four HDM snapshots where varying the parameters results in very different CO₂ saturation at a certain timestep. To further analyze the parameter domain range for porosity and permeability, we developed a ranking for the layers by converting the 2-D distributions into 1-D. To achieve this, the use of either harmonic, geometric or arithmetic averages as demonstrated by the equation below can facilitate that.

$$x_i^* = \left(\frac{1}{V} \int_V [x_i(y)]^{\omega_i} dV \right)^{\frac{1}{\omega_i}}$$

where x can be porosity or permeability, V_b is cell volume. Also, arithmetic, geometric, and harmonic averages correspond to $\omega = 1$, $\omega \rightarrow 0$, and $\omega = -1$, respectively. Moreover, the use of arithmetic averaging can usually provide values on the higher end. Using this 1-D distribution, we developed three porosity multipliers, which are 0.5, 1 and 1.5. Applying the porosity multipliers to each layer in the 3-D model along with 3 injection rates (Q) and 3 time-steps, resulted in 2,295 realizations.

Table 3—Summary of parameter ranges for sampling

Parameters	Min	Max
3 Porosity Multipliers	4.60×10^{-2}	0.338102
85 Permeability (m^2)	4.69×10^{-14}	1.16×10^{-12}
3 Injection Rates (kg/s)	0.001	0.01
3 Timesteps (years)	3.33	10.0

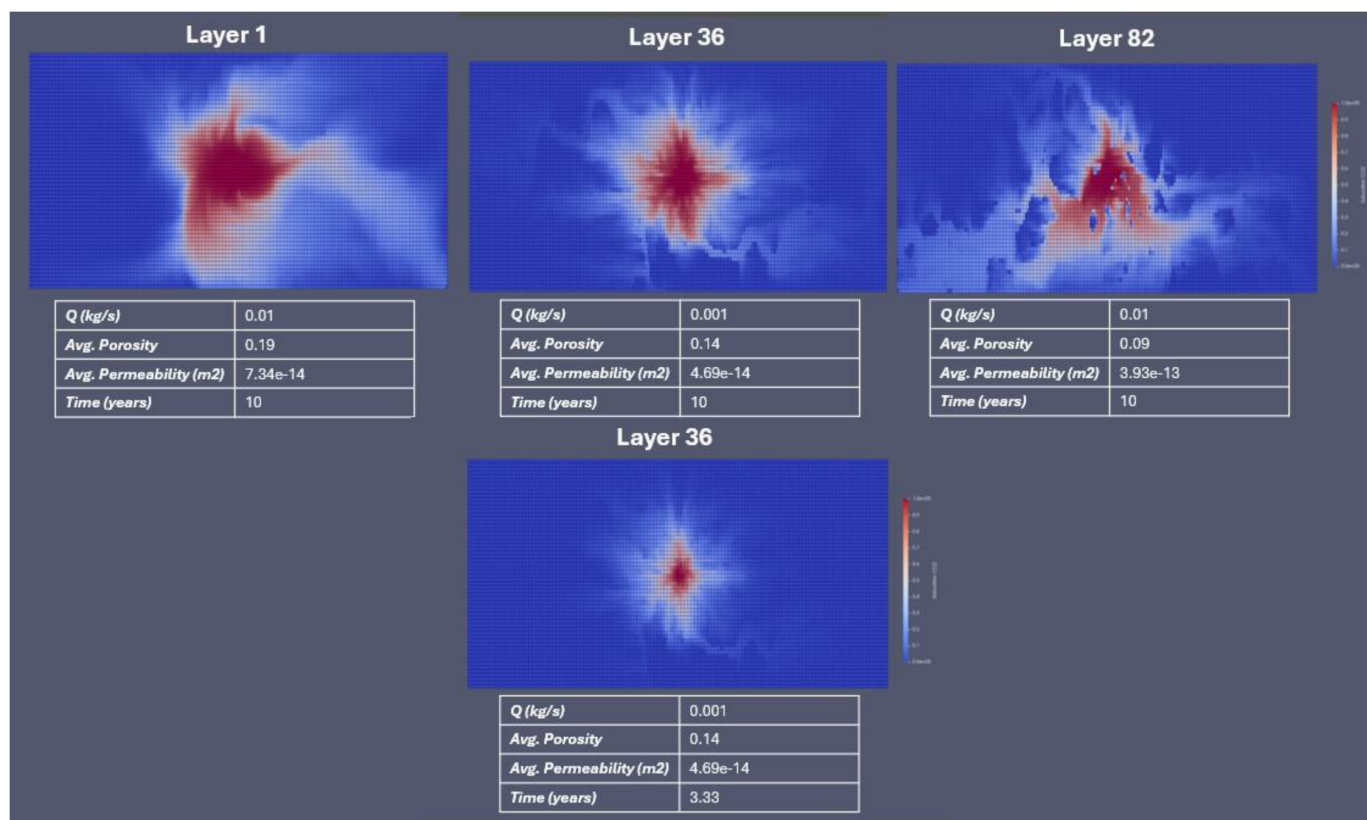


Figure 3—HDM solution snapshots showing the impact of varying the parameters on the shape of injected CO₂ plume.

Sampling Techniques

In this study, two non-adaptive sampling methods were implemented, which are random and Latin hypercube (LHS), in addition to one adaptive technique which is greedy sampling. These methods are discussed in-detail in [XXX]. Then the accuracy of the predicted ROMs using these techniques was compared via maximum absolute and average relative errors.

Projection-Based Reduced Order Model Description

As explained, the HDM simulation workflow allows to output the Jacobians and residuals for each newton iteration which were used to construct a PROM to model the pressure and saturation solutions in this study.

This PROM was a Petrov-Galerkin constructed using a Proper Orthogonal Decomposition (POD) on each newton iteration of the sampled HDM solutions. This allowed to generate a Reduced Order Basis (ROB) for each newton iteration. These ROB's were generated using singular value decomposition (SVD). In order to compare the PROM predictions with the HDM solutions, the predictions (δX) from each newton iteration are all added until the converged solution. An outline of the workflow is explained in Algorithm (1) in Figure 4. To ensure that random sampling will not result in some newton iterations with no sufficient data (i.e. no solution) to construct an ROB, we always include one sample with the highest number of newton iterations before starting the sampling with the different techniques. This results in an additional sample to the sample size collected using each technique, so the number of actual samples $k = 11$ for a 10-sample case.

Algorithm 1: Petrov-Galerkin PROM algorithm.

Selecting 10 Samples (using different techniques):

$$\delta X_{k \in (1, \dots, 10)}^{\nu \in (1, \dots, \nu^*)} \longrightarrow w_{k \in (1, \dots, 10)}^{\nu \in (1, \dots, \nu^*)}, \text{ where } \nu^* \text{ is the converged newton iteration.}$$
Generating ROB's:

Do the following in order to get ROB's for each newton iteration ν

for $\nu = 1, \dots, \nu^*$ **do**

A) Collect the sampled solutions into a matrix: $\mathbf{w} \in \mathbb{R}^{N \times k}$

for $k = 1, \dots, 10$ **do**

if w_k^ν *does not exist* (i.e. newton has converged at step $\nu - 1$) **then**

$\mathbf{w}(1 : N, k) = \emptyset$,

else

$\mathbf{w}(1 : N, k) = w_k^\nu$

end

B) Perform SVD:

$[\mathbf{U}, \sim, \sim] = \text{SVD}(\mathbf{w})$

C) Generate a Petrov-Galerkin ROB for ν :

$\mathbf{V} \longrightarrow \mathbf{U}$,

$\mathbf{W} \longrightarrow \mathbf{J}^\nu \times \mathbf{V}$,

end

Prediction and Error Estimation:

For the number of HDM solutions that have been generated S :

for $i = 1, \dots, S$ **do**

A) Generate prediction for each newton iteration ν :

for $\nu = 1, \dots, \nu^*$ **do**

if w_i^ν *does not exist* (i.e. newton has converged at step $\nu - 1$) **then**

$\tilde{w}_i^\nu \longrightarrow 0$,

else

$\mathbf{J}_{r,i}^\nu \longrightarrow \mathbf{W}^T \times \mathbf{J}_i^\nu \times \mathbf{V}$

$\mathbf{R}_{r,i}^\nu \longrightarrow \mathbf{W}^T \times \mathbf{R}_i^\nu$

$q_i \longrightarrow -\mathbf{J}_{r,i}^\nu / \mathbf{R}_{r,i}^\nu$

$\tilde{w}_i^\nu \longrightarrow \mathbf{V} \times q_i$

end

B) Compute the final converged prediction from individual newton iterations:

$\tilde{w}_i = \sum \tilde{w}_i^\nu$

C) Estimate the error for the sample i :

$\epsilon_i \longrightarrow \|(\tilde{w}_i - w_i)\|_2 / \|w_i\|_2$,

end

Figure 4—Petrov-Galerkin PROM Algorithm.

Results

This workflow worked well for the top (Tarbert) layers, and the prediction accuracy for the 10-sample PROM application was within a margin of 5% for the maximum error in all grid cells and a mean error

of less than 1%. Here we showcase the capabilities of PROM in predicting the first time snapshot only. Different sampling techniques resulted in different accuracies as shown in the table below:

Table 4—Comparison of different sampling techniques for $k = 11$.

Error	Random Sampling	LHS Sampling	Greedy Sampling
Maximum	4.9×10^{-2}	4.0×10^{-2}	2.7×10^{-2}
Mean	8.2×10^{-3}	7.8×10^{-3}	6.0×10^{-3}

Table 5—Comparison of different sampling techniques to achieve a maximum error less than 5%

Optimized	Random Sampling	LHS Sampling	Greedy Sampling
Maximum Error	4.9×10^{-2}	4.8×10^{-2}	4.4×10^{-2}
Samples (k)	11	9	5

The total number of predictions are 315 from those top 35 layers by varying porosity multipliers and injection rates. On the other hand, the remaining bottom (Ness) 50 layers showed poor prediction in general (unless these layers are used in the sampled data which will grow k wildly) except for a few layers. This result was expected since high-permeability channels in Ness formation result in a more pronounced CO_2 viscous-dominated flow, which can produce a shock at the CO_2 front, compared to Tarbert, where the flow is gravity- and/or capillary-dominated where any shock that can be present will become more diffused. A summary of the potential measures that can be taken to improve such results are discussed in the next section.

Conclusions

In conclusion, the results highlight the efficiency of the linear Petrov-Galerkin method in reducing computational complexity for the upper isotropic and heterogeneous layers of the model, achieving less than 5% error with as few as 11 non-adaptive samples, and further improving efficiency to just five samples through adaptive sampling. However, the increased geological complexity and anisotropy in the lower layers underscore the need for more advanced nonlinear model order reduction techniques to ensure accurate representation of their influence on reservoir performance. These findings emphasize the importance of tailoring model reduction strategies to the underlying geological characteristics to achieve both accuracy and computational efficiency with the following potential future extensions:

- Investigate sampling accuracy on the remaining timesteps by extending PROM to affine approximation formulation and using the solution from the first timestep as an initial condition to the second timestep prediction and similarly for the third timestep.
- Improve sampling accuracy by:
 - Implementing local ROBs for Tarbet and Ness formations to capture the rich parameter space (Figures 1 and 2) [Amsallem et al., 2012].
 - Implementing non-linear approximation methods to capture flow shocks generated by the two-phase transport equation especially due to the permeability channels in Ness formation such as the quadratic approximation method to mitigate Kolmogorov barrier [Barnett and Farhat, 2022].
 - Simplify the flow and transport problem for the Ness layers by simulating a single-phase flow as a starting point and predict the pressure solution only before extending to the two-phase flow.
- Extend this workflow to highly nonlinear problems involving all physical processes that are relevant to CO_2 injection in saline aquifers. This includes miscibility of the gas into the liquid phase

and vice versa, chemical dissolution of salts and rock minerals, capillary trapping due to hysteresis effects, and geomechanics.

References

- Alali, A. (2023). Modeling co2 geologic storage using machine learning. In *SPE Middle East Oil and Gas Show and Conference*, page D021S066R004. SPE.
- Amsallem, D., Zahr, M.J., and Farhat, C. (2012). Nonlinear model order reduction based on local reduced-order bases. *International Journal for Numerical Methods in Engineering*, **92**(10): 891–916.
- Aziz, K. and Settari, A. (1979). Petroleum reservoir simulation, volume **476**. *Applied Science Publishers London*.
- Barnett, J. and Farhat, C. (2022). Quadratic approximation manifold for mitigating the kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, **464**: 111348.
- Bazargan, H., Christie, M., Elsheikh, A.H., and Ahmadi, M. (2015). Surrogate accelerated sampling of reservoir models with complex structures using sparse polynomial chaos expansion. *Advances in Water Resources*, **86**: 385–399.
- BP, B.P. (2021). Statistical review of world energy 2022. London: BP.
- Bruant, R., Guswa, A., Celia, M., and Peters, C. (2002). Safe storage of co2 in deep saline aquifers. *ENVIRONMENTAL SCIENCE AND TECHNOLOGY-WASHINGTON DC* **36**(11): 240A–245A.
- Cardoso, M.A., Durlowsky, L.J., and Sarma, P. (2009). Development and application of reduced-order modeling procedures for subsurface flow simulation. *International journal for numerical methods in engineering*, **77**(9): 1322–1350.
- Cook, J., Nuccitelli, D., Green, S.A., Richardson, M., Winkler, B., Painting, R., Way, R., Jacobs, P., and Skuce, A. (2013). Quantifying the consensus on anthropogenic global warming in the scientific literature. *Environmental research letters*, **8**(2): 024024.
- Farnsworth, S.J. and Lichter, S.R. (2012). The structure of scientific opinion on climate change. *International Journal of Public Opinion Research*, **24**(1): 93–103.
- Hamdi, H., Couckuyt, I., Sousa, M.C., and Dhaene, T. (2017). Gaussian processes for history-matching: application to an unconventional gas reservoir. *Computational Geosciences*, **21**: 267–287.
- Hitchon, B., Gunter, W., Gentzis, T., and Bailey, R. (1999). Sedimentary basins and greenhouse gases: a serendipitous association. *Energy Conversion and Management*, **40**(8): 825–843.
- Hoffert, M.I., Caldeira, K., Jain, A.K., Haites, E.F., Harvey, L.D., Potter, S.D., Schlesinger, M.E., Schneider, S.H., Watts, R.G., Wigley, T.M., et al (1998). Energy implications of future stabilization of atmospheric co2 content. *Nature*, **395**(6705): 881–884.
- Houghton, J.T. (1996). Climate change 1995: The science of climate change: contribution of working group I to the second assessment report of the Intergovernmental Panel on Climate Change, volume **2**. Cambridge University Press.
- Ju, X., Hamon, F.P., Wen, G., Kanfar, R., Araya-Polo, M., and Tchelepi, H.A. (2024). Learning co2 plume migration in faulted reservoirs with graph neural networks. *Computers Geosciences*, **193**: 105711.
- Metz, B., Davidson, O., De Coninck, H., Loos, M., and Meyer, L. (2005). IPCC special report on carbon dioxide capture and storage. Cambridge: Cambridge University Press.
- Myers, K.F., Doran, P.T., Cook, J., Kotcher, J.E., and Myers, T.A. (2021). Consensus revisited: Quantifying scientific agreement on climate change and climate expertise among earth scientists 10 years later. *Environmental Research Letters*, **16**(10): 104030.
- Oreskes, N. (2004). The scientific consensus on climate change. *Science*, **306**(5702): 1686–1686.
- Orr, F.M.Jr (2004). Storage of carbon dioxide in geologic formations. *Journal of Petroleum Technology*, **56**(09): 90–97.
- Pruess, K., Oldenburg, C.M., and Moridis, G. (1999). Tough2 user's guide version.
- Pruess, K., Xu, T., Apps, J., and Garcia, J. (2003). Numerical modeling of aquifer disposal of co2. *Spe Journal*, **8**(01): 49–60.
- Razavi, S., Tolson, B.A., and Burn, D.H. (2012). Review of surrogate modeling in water resources. *Water Resources Research*, **48**(7).
- Settgast, R.R., Aronson, R.M., Besset, J.R., Borio, A., Bui, Q.M., Byer, T.J., Castelletto, N., Citrain, A., Corbett, B.C., Corbett, J., Cordier, P., Cremon, M.A., Crook, C.M., Cusini, M., Fei, F., Frambati, S., Franc, J., Franceschini, A., Frigo, M., Fu, P., Gazzola, T., Gross, H., Hamon, F., Han, B.M., Hao, Y., Hasanzade, R., Homel, M., Huang, J., Jin, T., Ju, I., Kachuma, D., Karimi-Fard, M., Kim, T., Klevtsov, S., Lapene, A., Magri, V.A.P., Mazuyer, A., N'diaye, M., Osei-Kuffuor, D., Povolny, S., Ren, G., Semnani, S.J., Sherman, C.S., Rey, M., Tchelepi, H.A., Tobin, W.R., Tomin, P., Untereiner, L., Vargas, A., Waziri, S., Wen, X., White, J.A., and Wu, H. (2024). GEOS: A performance portable multi-physics simulation framework for subsurface applications. *Journal of Open Source Software*, **9**(102): 6973.
- Tian, L., Wilkinson, R., Yang, Z., Power, H., Fagerlund, F., and Niemi, A. (2017). Gaussian process emulators for quantifying uncertainty in co2 spreading predictions in heterogeneous media. *Computers & Geosciences*, **105**: 113–119.

- Townsend, A., Raji, N., and Zapantis, A. (2020). The value of carbon capture and storage (ccs). Global CCS Institute: Docklands, Australia.
- Wen, G., Li, Z., Long, Q., Azizzadenesheli, K., Anandkumar, A., and Benson, S.M. (2023). Real-time high-resolution co₂ geological storage prediction using nested fourier neural operators. *Energy & Environmental Science*, **16**(4): 1732–1741.
- Wigley, T.M., Richels, R., and Edmonds, J.A. (1996). Economic and environmental choices in the stabilization of atmospheric co₂ concentrations. *Nature*, **379**(6562): 240–243.
- Zhai, P., Portner, H., Roberts, D., Skea, J., Shukla, P., Pirani, A., Moufouma-Okia, W., Pean, C., Pidcock, R., Connors, S., et al (2018). Global warming of 1.5 c. an ipcc special report on the impacts of global warming of 1.5 c above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change. *Sustainable development, and efforts to eradicate poverty*, page 32.